

Q-CARE

Quilvian Care Reminder & Engagement

Developer Handoff & Implementation Blueprint

Submodule of Quilvian Patient Journey

Document Type	Technical Product Specification / Developer Handoff
Version	1.0
Status	Ready for Technical Breakdown
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Primary Channel	WhatsApp Business API - extensible to omnichannel

Purpose: satu engine reminder terpusat untuk perjalanan klinis, administratif, dan finansial pasien, termasuk respons pasien dan tindak lanjut petugas.

1. Nama Modul dan Positioning

1.1 Nama yang Direkomendasikan

Q-CARE - Quilvian Care Reminder & Engagement

Nama ini direkomendasikan karena fungsi sistem tidak berhenti pada mengirim reminder. Q-CARE juga menerima respons pasien, memperbarui Patient Journey, membuat task internal, melakukan eskalasi, dan memastikan suatu tindakan benar-benar selesai.

1.2 Posisi di Ekosistem Quilvian

Lapisan	Peran Q-CARE
Quilvian Patient Journey	Q-CARE menjadi automation engine yang menempel pada setiap milestone perjalanan pasien.
Clinical Systems	Menerima event dari appointment, LIS, RIS/PACS, farmasi, dan EMR.
Financial Systems	Menerima event invoice, pembayaran, aging, dan reimbursement.
Patient Engagement	Mengirim WA, menerima respons, deep link, upload bukti, dan konfirmasi.
Operations	Membuat task, worklist, escalation, dashboard, dan audit trail.

1.3 Nama Alternatif

Nama	Kelebihan	Catatan
Quilvian Journey Reminder Engine (QJRE)	Sangat teknis dan jelas sebagai engine.	Kurang menggambarkan engagement dan follow-up.
Quilvian Patient Engagement Hub	Cocok untuk pengembangan omnichannel.	Terlalu luas untuk scope awal.
Quilvian Smart Reminder	Mudah dipahami user.	Kurang kuat sebagai nama arsitektur developer.

2. Ringkasan Fungsi

Q-CARE adalah centralized event-driven reminder and engagement engine. Sistem menerima event dari modul sumber, mengevaluasi rule, menjadwalkan pesan, mengirim WhatsApp, membaca respons pasien, memperbarui status perjalanan, dan membuat tindak lanjut internal jika dibutuhkan.

- Reminder laboratorium: jadwal, persiapan, belum datang, dan hasil tersedia.
- Reminder radiologi: jadwal, persiapan, konfirmasi hadir, no-show, dan hasil tersedia.

- Reminder kontrol: H-7/H-3/H-1, konfirmasi, batal, reschedule, dan no-show follow-up.
- Reminder obat: jadwal minum, obat hampir habis, refill, belum diambil, dan keluhan.
- Reminder pembayaran: invoice, due date, outstanding, partial payment, overdue, dan payment link.
- Reminder reimbursement: dokumen oleh RS/pasien/hybrid, status klaim, estimasi dana cair, dan pembayaran pasien ke RS.
- Task & escalation: kegagalan pesan, non-response, risiko klinis, no-show, overdue, atau exception.
- Dashboard & audit: delivery/read/response, conversion, completion, SLA, aging, dan histori end-to-end.

3. Alur End-to-End

Step	Stage	Behavior
1	Source Event	SIMRS/LIS/RIS/Farmasi/Billing/Reimbursement menghasilkan event.
2	Event Ingestion	Q-CARE menerima event dan melakukan validation + deduplication.
3	Rule Evaluation	Rules Engine mencari rule aktif berdasarkan kategori, unit, tanggal, status, penjamin, dan konteks.
4	Reminder Scheduling	Scheduler membuat reminder instance dan jadwal kirim.
5	Pre-send Validation	Cek consent, nomor WA, quiet hour, current status, stop condition, dan duplicate lock.
6	Message Delivery	Messaging adapter mengirim template melalui WhatsApp Business API.
7	Delivery Tracking	Webhook memperbarui Sent -> Delivered -> Read -> Responded atau Failed.
8	Patient Action	Pasien memilih confirm/reschedule/sudah minum/sudah bayar/belum menerima dana/upload bukti/butuh bantuan.
9	Journey Update	Response Handler memperbarui journey item dan memanggil service terkait bila diizinkan.
10	Task/Escalation	Jika action tidak selesai, response ambigu, atau SLA lewat, sistem membuat task internal.
11	Stop/Close	Reminder berhenti ketika target action selesai, dibatalkan, disuppress, atau closed.

Step	Stage	Behavior
12	Analytics	Semua event masuk dashboard, audit trail, dan reporting.

Golden Rule

Q-CARE tidak boleh menjadi kumpulan cron job per fitur. Semua domain harus memakai satu Reminder Engine, satu status model, satu mekanisme scheduler/queue, satu audit trail, dan adapter domain yang terpisah.

4. Komponen Sistem yang Harus Dibangun

Component	Responsibility
Event Gateway	Menerima domain event dari sistem lain. Validasi schema, authentication, correlation ID, idempotency.
Rules Engine	Menentukan apakah reminder dibuat, kapan dikirim, kepada siapa, template apa, stop condition, retry, dan escalation.
Scheduler Service	Mengubah rule + reference date menjadi due schedule. Mendukung H-x, H+x, jam, recurring, business day, quiet hour.
Reminder Service	Lifecycle reminder instance, status, cancellation, suppression, reschedule, manual override.
Template Service	Template, language, variables, versioning, preview, approval state, provider template ID.
WhatsApp Adapter	Outbound send, callback verification, delivery status, inbound button/text/media, provider error mapping.
Response Router	Mapping respons ke intent/action dan domain handler.
Task & Escalation Service	Worklist internal, assignment, SLA, severity, escalation level, resolution.
Consent & Contact Service	Validasi nomor, preferred language, opt-in/opt-out, authorized family contact.
Domain Adapters	Appointment, Lab, Radiology, Medication, Billing, Reimbursement.
Dashboard API	Metrics, filters, operational queue, drill-down, health monitoring.
Audit & Observability	Audit events, logs, traces, metrics, dead-letter queue, reconciliation.

5. Domain Flow Detail

5.1 Laboratorium

Trigger	Default Action	Patient CTA	Stop Condition
Lab order + schedule	Create reminder H-1 or configured offset with preparation instructions.	Confirm / Need help	Specimen received, order cancelled, schedule changed.
Not arrived at due time	Send missed-lab reminder and optionally task.	Reschedule / Contact us	New schedule or test completed.
Validated result available	Notify result availability only; do not expose sensitive detail in message preview.	Open secure portal / View instruction	Notification delivered or case closed.

5.2 Radiologi

Trigger	Default Action	Patient CTA	Stop Condition
Radiology appointment	H-3 and H-1 reminder with modality-specific preparation.	Confirm / Reschedule	Exam completed/cancelled.
No-show	Post-appointment follow-up.	Reschedule / Need help	New appointment or closure.
Validated result	Notify availability via secure link.	Open portal	Delivered/closed.

5.3 Kontrol / Appointment

Stage	Default Timing	Patient CTA	System Behavior
Pre-appointment	H-7	Confirm / Reschedule / Cancel	Record response; open reschedule task if needed.
Pre-appointment	H-3	Confirm / Reschedule	Suppress if already cancelled or completed.
Pre-appointment	H-1	Confirm / Directions	Last standard reminder.
No-show	H+1	Reschedule	Create customer-care worklist if no action.

5.4 Obat

Trigger	Behavior	Patient CTA	Escalation
Medication schedule	Send dose reminder according to verified medication plan.	Sudah diminum / Belum / Lewati / Ada keluhan	Keluhan creates clinical task.
Estimated stock low	Reminder refill/pick-up.	Ajukan refill / Need help	Pharmacy task if required.

Trigger	Behavior	Patient CTA	Escalation
Medication discontinued	Cancel future reminder instances.	None	No escalation.

Clinical Safety Boundary

Q-CARE hanya mengingatkan berdasarkan medication plan yang sudah disahkan sistem klinis. Q-CARE tidak menghitung dosis, mengubah terapi, atau memberikan diagnosis.

5.5 Pembayaran

Trigger	Behavior	Patient CTA	Stop Condition
Invoice issued	Send invoice notification if configured.	Pay now / View invoice	Paid/cancelled.
Approaching due date	Send H-3/H-1 reminder using current outstanding amount.	Pay now / Need help	Paid or due date updated.
Partial payment	Recalculate remaining balance; next message uses latest outstanding.	Pay remaining	Outstanding zero.
Overdue	Run H+1/H+3/H+7/H+14 or configured sequence.	Pay / Upload proof / Need help	Paid, dispute, waiver.

5.6 Reimbursement

Model	Flow	Reminder Target	Completion
Hospital submits	RS prepares -> sends docs -> insurer review -> paid to patient -> patient pays RS.	Internal officer for docs; patient for settlement.	Hospital invoice fully settled.
Patient submits	RS hands over package -> patient sends -> upload proof -> insurer review -> paid to patient -> patient pays RS.	Patient for submission and settlement.	Hospital invoice fully settled.
Hybrid	Each document has owner Hospital or Patient. Submission waits for mandatory items.	Per-document owner.	All mandatory docs submitted, then settlement complete.

6. Reminder Rules Engine

Rule adalah konfigurasi bisnis yang dapat dibuat admin. Rule harus versioned, memiliki effective date, dan tidak di-hardcode di service domain.

Field	Type	Example / Notes
rule_id	UUID	Unique identifier.
name	string	CONTROL_H1_DEFAULT.
category	enum	LAB, RADIOLOGY, CONTROL, MEDICATION, PAYMENT, REIMBURSEMENT.
trigger_event	string	appointment.created, invoice.overdue, reimbursement.paid_to_patient.
condition_json	JSON	unit, insurer, service type, status, min amount, patient segment.
reference_time_field	string	appointment_at / due_date / expected_payment_at.
offset_minutes	integer	-1440 = H-1; +4320 = H+3.
recurrence	JSON/null	Repeat every N hours/days, max_count.
quiet_hour_policy	ref	Default 20:00-07:00 or hospital config.
template_id	UUID	Template version resolved at scheduling/send time.
recipient_policy	enum	PATIENT, GUARANTOR, INTERNAL_USER.
stop_condition_json	JSON	paid, completed, cancelled, specimen_received.
retry_policy	JSON	max retries, delay, error classes.
escalation_policy_id	UUID	Task generation and escalation chain.
priority	enum	LOW, NORMAL, HIGH, URGENT.
effective_from/to	datetime	Version validity.
is_active	boolean	Administrative switch.

```
{
  "name": "CONTROL_H1_DEFAULT",
  "category": "CONTROL",
  "trigger_event": "appointment.confirmed",
  "reference_time_field": "appointment_at",
  "offset_minutes": -1440,
  "conditions": {"status": ["BOOKED", "CONFIRMED"]},
  "template_code": "CONTROL_REMINDER_H1_ID",
  "stop_conditions": ["COMPLETED", "CANCELLED"],
  "recipient": "PATIENT",
  "priority": "NORMAL"
}
```

7. Status Model dan State Transition

Status	Meaning	Allowed Next
SCHEDULED	Reminder sudah dibuat dan menunggu due time.	QUEUED, CANCELLED, SUPPRESSED.
QUEUED	Masuk message queue untuk dikirim.	SENT, FAILED, CANCELLED.
SENT	Provider menerima message request.	DELIVERED, FAILED.
DELIVERED	Pesan sampai ke device/provider destination.	READ, RESPONDED, EXPIRED.
READ	Pesan dibaca.	RESPONDED, EXPIRED.
RESPONDED	Respons pasien diterima dan diproses.	COMPLETED, ESCALATED.
FAILED	Pengiriman gagal.	QUEUED retry, ESCALATED, CLOSED.
SUPPRESSED	Tidak dikirim karena consent/quiet rule/invalid state.	SCHEDULED if resumed, CLOSED.
CANCELLED	Tidak lagi relevan.	Final.
EXPIRED	Window respons berakhir.	ESCALATED, CLOSED.
COMPLETED	Target action selesai.	Final.

Idempotency

Gunakan idempotency key minimal: patient_id + source_system + source_reference + rule_version + occurrence_time. Event replay tidak boleh menciptakan reminder ganda.

8. Event Catalog Minimum

Event	Source	Purpose
patient.contact.updated	SIMRS	Nomor WA / consent berubah.
appointment.created	Appointment	Jadwal baru dibuat.
appointment.rescheduled	Appointment	Tanggal/jam berubah.
appointment.cancelled	Appointment	Jadwal dibatalkan.
appointment.no_show	Appointment	Pasien tidak hadir.
lab.order.scheduled	LIS	Pemeriksaan lab terjadwal.
lab.specimen.received	LIS	Stop future pre-test reminders.
lab.result.validated	LIS	Hasil tersedia untuk notifikasi.
radiology.order.scheduled	RIS	Radiologi terjadwal.

Event	Source	Purpose
radiology.exam.completed	RIS	Stop schedule reminder.
radiology.result.validated	RIS	Hasil tersedia.
medication.plan.activated	EMR/Farmasi	Mulai medication reminder.
medication.plan.changed	EMR/Farmasi	Reschedule future doses.
medication.plan.stopped	EMR/Farmasi	Cancel future doses.
invoice.issued	Billing	Invoice aktif.
invoice.partial_paid	Billing	Outstanding berubah.
invoice.paid	Billing	Stop all payment reminders.
invoice.overdue	Billing	Start collection reminder path.
reimbursement.case.created	RJM	Workflow dimulai.
reimbursement.document.pending	RJM	Reminder ke owner dokumen.
reimbursement.submitted	RJM	Stop document reminders.
reimbursement.paid_to_patient	RJM	Start payment-to-hospital path.
reimbursement.closed	RJM	Stop all reimbursement reminders.

9. Data Model Minimum

Entity	Purpose
reminder_rule	Master rule + version + effective dates.
reminder_template	Message body, variables, language, provider template ID, approval status, version.
reminder_instance	One scheduled reminder occurrence linked to patient + domain source.
reminder_delivery	Every outbound attempt and provider status callback.
patient_response	Inbound button/text/media + parsed intent + processing status.
journey_event	Normalized event stored for replay/audit/reconciliation.
patient_contact_preference	WA number, language, consent, opt-out, authorized contact.
followup_task	Internal work item, owner, priority, SLA, status, resolution.
escalation_policy	Thresholds and target roles.

Entity	Purpose
domain_reference	Reference to appointment/order/invoice/reimbursement case.
attachment_evidence	Proof of transfer/submission and secure storage reference.
audit_log	Immutable create/update/override/security audit.
outbox_event	Transactional outbox for reliable event publication.
dead_letter	Failed events/messages after retry limit.

10. API Contract - Minimum Endpoints

Method	Endpoint	Function
POST	/api/v1/events	Ingest normalized domain event. Internal only.
GET	/api/v1/reminders/{id}	Get reminder detail, history, delivery, response.
POST	/api/v1/reminders/{id}/cancel	Cancel with reason and actor.
POST	/api/v1/reminders/{id}/reschedule	Move scheduled time with reason.
POST	/api/v1/reminders/{id}/send-now	Privileged manual send.
GET	/api/v1/rules	List/filter active rule versions.
POST	/api/v1/rules	Create rule draft.
POST	/api/v1/rules/{id}/activate	Activate version with effective date.
GET	/api/v1/templates	List templates and provider status.
POST	/api/v1/templates	Create/update draft template.
POST	/webhooks/whatsapp/status	Receive send/delivery/read/failure callbacks.
POST	/webhooks/whatsapp/inbound	Receive patient messages/buttons/media.
GET	/api/v1/tasks	Operational worklist.
POST	/api/v1/tasks/{id}/resolve	Resolve task with outcome.
GET	/api/v1/dashboard/summary	Dashboard aggregates with filters.
GET	/api/v1/audit	Privileged audit query.

```
POST /api/v1/events
{
  "event_id": "evt_01J...",
```

```

"event_type": "invoice.overdue",
"occurred_at": "2026-08-12T10:00:00+07:00",
"source": "billing",
"correlation_id": "corr_123",
"patient_id": "P000123",
"source_reference": "INV-2026-000991",
"payload": {
  "invoice_number": "INV-2026-000991",
  "outstanding_amount": 2500000,
  "due_date": "2026-08-11"
}
}

```

11. Scheduler, Queue, Retry, dan Consistency

- Event ingestion harus durable. Simpan event sebelum diproses atau gunakan transactional outbox/inbox pattern.
- Reminder instance dibuat idempotent. Duplicate event harus menghasilkan no-op, bukan pesan ganda.
- Scheduler hanya memindahkan reminder due ke queue; worker yang melakukan send.
- Sebelum send, worker wajib re-check stop condition dan current domain state.
- Provider callback diproses idempotent berdasarkan provider_message_id + callback status + provider timestamp.
- Retry hanya untuk error teknis yang retryable. Validation error, opt-out, dan permanent provider rejection tidak di-retry.
- Dead-letter queue wajib memiliki UI/operational endpoint untuk replay setelah masalah diselesaikan.
- Payment callback dan clinical completion event memiliki prioritas tinggi untuk menghentikan reminder yang masih menunggu.
- Semua worker harus stateless; state disimpan pada database/queue yang disepakati.
- Time-based schedule menggunakan Asia/Jakarta sebagai display/default business timezone, tetapi timestamp persistence direkomendasikan konsisten dalam UTC.

12. WhatsApp Messaging Behavior

Area	Requirement
Template	Gunakan provider-approved template untuk outbound proactive messaging.
Variables	Render dari trusted fields; validate required variables before queue.
Sensitive Data	Jangan menampilkan diagnosis/detail hasil klinis pada preview message. Gunakan secure link.
Buttons	Gunakan structured CTA untuk confirm, reschedule, paid, not paid, upload proof, need help.
Inbound Text	Parse ke intent; jika ambiguous, create task for human review.

Area	Requirement
Media	Evidence upload disimpan ke secure object storage/DMS; message DB hanya menyimpan reference.
Fallback	Jika WA gagal permanen, buat internal task; channel lain dapat ditambah pada fase berikutnya.
Quiet Hours	Default business rule dapat dikonfigurasi; urgent messages harus memiliki policy terpisah.
Rate Limit	Adapter harus menghormati provider limit dan menggunakan queue backpressure.

13. Patient Response Mapping

Intent	Domain	Action
CONFIRM_ATTENDANCE	Appointment	Set patient confirmation; no further pre-appointment confirmation required unless rule says otherwise.
REQUEST_RESCHEDULE	Appointment	Create reschedule task or call scheduling API when allowed.
CANCEL_APPOINTMENT	Appointment	Request cancellation workflow; do not directly cancel if approval required.
MED_TAKEN	Medication	Store adherence event.
MED_NOT_TAKEN	Medication	Store adherence event; optional follow-up based on rule.
MED_COMPLAINT	Medication	Create high-priority clinical task.
PAY_NOW	Billing	Generate/open secure payment flow.
PAYMENT_PROOF	Billing/RJM	Store evidence and create verification task.
INSURER_NOT_PAID	RJM	Pause collection path as configured; move case to dispute/waiting insurer payment.
DOCUMENT_SENT	RJM	Request proof; after verification stop document reminder.
NEED_HELP	Any	Create customer care task.
UNRECOGNIZED	Any	Route to manual inbox/task.

14. Task and Escalation

Condition	Default Task Owner	Priority	Example SLA
WA invalid/permanent failure	Customer Care	NORMAL	1 business day
Patient asks reschedule	Appointment team	NORMAL	4 working hours
Medication complaint	Nurse/Pharmacy/Clinical team	HIGH	According to clinical policy
Lab/Radiology no-show	Relevant unit / Customer Care	NORMAL	Same/next business day
Payment overdue > threshold	Collection Officer	HIGH	According to aging policy
Reimbursement proof waiting verification	Insurance/Finance	NORMAL	1 business day
Broken promise-to-pay	Collection Supervisor	HIGH	Same business day
Critical system backlog	IT/DevOps	URGENT	Immediate operational response

15. Dashboard dan KPI

Dashboard	Minimum Metrics
Operational	Scheduled, queued, sent, delivered, read, responded, failed, suppressed, cancelled, expired.
Journey Completion	Completed action per category, no-show, reschedule, refill, document submitted, paid.
Financial	Invoice outstanding, collected after reminder, partial payment, aging, reimbursement outstanding.
Engagement	Delivery rate, read rate, response rate, CTA conversion, time-to-response.
SLA & Escalation	Tasks due/overdue, SLA breach, escalation level, owner workload.
System Health	Queue depth, oldest queued message, callback lag, retry count, provider error rate, DLQ volume.

16. Role-Based Access Control

Role	View	Change
System Admin	All configuration and system health.	Rules, templates, roles, technical config; no business settlement override unless granted.
Unit Operator	Relevant patient reminder/worklist.	Operational status and task resolution.

Role	View	Change
Clinical User	Clinical-related task and journey data.	Clinical follow-up outcome only.
Finance/Collection	Billing/reimbursement journey.	Payment verification, collection outcome, dispute flags.
Manager	Aggregated dashboards and permitted drill-down.	Selected override/approval actions.
Auditor	Audit and reports.	Read only.

17. Non-Functional Requirements

Area	Requirement
Availability	Target minimum 99.5% monthly excluding approved maintenance.
Performance	Event-to-reminder creation ≤ 60 sec; final-state cancellation ≤ 5 min under normal load.
Scale	Design initial capacity for $\geq 100,000$ reminder instances/day, horizontally scalable.
Security	TLS, encryption at rest, RBAC, MFA for privileged admin, secret vault, signed/verified webhooks.
Privacy	Minimum necessary data in messages/logs; secure links with expiry; consent and access control.
Reliability	Durable queue, retry with backoff, idempotency, DLQ, reconciliation.
Observability	Metrics, logs, distributed correlation ID, alerting, provider health dashboard.
Audit	Immutable history for configuration, manual override, outbound/inbound events, and payment evidence.
Deployability	Separate DEV/UAT/PROD; migration scripts; rollback plan; feature flags.
Documentation	OpenAPI/event schema, runbook, configuration guide, UAT checklist, release notes.

18. Error Handling and Edge Cases

- Patient has no valid WhatsApp number -> suppress send and create verification task if category requires follow-up.
- Patient changes number -> future reminders use latest verified contact; historical deliveries remain immutable.

- Appointment rescheduled after reminder scheduled -> cancel old instance and create schedule from new date.
- Invoice paid while message is queued -> worker re-checks outstanding and cancels send.
- Duplicate payment callback -> idempotent no-op.
- Provider callback arrives out of order -> state transition must not regress from READ to DELIVERED.
- Template disabled/rejected -> do not send; alert admin and route to configured fallback.
- Patient response after reminder expired -> store response; route through current domain state before action.
- System source unavailable -> retain event/reminder state and reconcile when source returns.
- Reimbursement patient states insurer has not paid -> pause collection escalation according to business rule; create insurer follow-up state/task.
- Medication regimen changed -> cancel all future dose reminder instances derived from old version and rebuild schedule.
- Multiple invoices/cases -> CTA must be bound to exact source reference, never infer from patient alone.

19. Development Breakdown - Epics

Epic	Name	Scope
EPIC-01	Core Domain + DB	Reminder rules, template, instance, delivery, response, task, audit entities and migrations.
EPIC-02	Event Ingestion	Event endpoint, schema validation, idempotency, outbox/inbox, correlation ID.
EPIC-03	Rules + Scheduler	Rule evaluation, offsets, recurrence, quiet hours, stop conditions, versioning.
EPIC-04	WhatsApp Adapter	Outbound template send, webhook status, inbound button/text/media, retry mapping.
EPIC-05	Response Router	Intent normalization and domain action dispatcher.
EPIC-06	Task/Escalation	Worklist, assignment, SLA, priority, escalation chain.
EPIC-07	Appointment Adapter	Control/radiology/lab appointment-related events and actions.
EPIC-08	Lab/Radiology Adapter	Order/result events and completion stop conditions.
EPIC-09	Medication Adapter	Medication plan schedule, change, stop, adherence response.
EPIC-10	Billing Adapter	Invoice, outstanding, payment callback, partial payment, overdue.

Epic	Name	Scope
EPIC-11	Reimbursement Adapter	Hospital/patient/hybrid document workflow + paid-to-patient settlement reminder.
EPIC-12	Admin UI	Rules, templates, escalation policy, preview, activate/version.
EPIC-13	Operational UI	Patient timeline, reminder detail, inbox, worklist, manual action.
EPIC-14	Dashboard	Operational, engagement, finance, SLA, system health.
EPIC-15	Hardening	Security, load, reconciliation, audit, DR, runbook, monitoring.

20. Codex Implementation Blueprint

Bagian ini dapat dijadikan work package untuk tim developer yang menggunakan Codex. Implementasi sebaiknya dilakukan incremental dan setiap epic memiliki contract test sebelum domain berikutnya ditambahkan.

Recommended Build Order

1) Core schema + status machine -> 2) Event ingestion -> 3) Rules/Scheduler -> 4) WA Adapter -> 5) Response Router -> 6) Task/Escalation -> 7) Domain adapters -> 8) Admin/Operational UI -> 9) Dashboard -> 10) Hardening/UAT.

20.1 Repository/Module Boundary

```

qcare/
domain/
  reminder/
  rule/
  template/
  response/
  task/
  consent/
application/
  event_ingestion/
  scheduling/
  delivery/
  response_routing/
  escalation/
adapters/
  whatsapp/
  appointment/
  lab/
  radiology/
  medication/
  billing/
  reimbursement/
api/

```


workers/
 persistence/
 observability/
 tests/
 unit/
 contract/
 integration/
 e2e/

20.2 Backend Rules

- Business logic must live in domain/application layer, not controller or provider adapter.
- Provider-specific payloads must not leak into reminder domain model.
- All external callbacks must be authenticated and idempotent.
- Database transaction creating/updating business state should publish via transactional outbox where asynchronous events are needed.
- Do not schedule one cron per patient. Use due_at indexed queue/scheduler strategy.
- All state transitions should go through a central transition validator to prevent invalid regression.
- All manual override endpoints require actor, reason, permission, and audit event.

20.3 Frontend Rules

- Build reusable status badge and timeline components shared across categories.
- Admin rule form must preview generated schedule and show stop/escalation behavior before activation.
- Operational worklist must support filter, sort, assignment, bulk safe actions, and drill-down.
- Patient timeline must visually distinguish system event, outbound reminder, inbound response, task, and payment/document event.
- Never show sensitive patient data in generic toast/log messages.

20.4 Required Automated Tests

- Unit tests for rule offset, recurrence, quiet hours, stop condition, state transitions.
- Contract tests for every inbound/outbound integration schema.
- Idempotency tests for duplicate events, duplicate callbacks, and retry.
- Time tests around midnight, timezone, DST-independent behavior, holidays/business day if enabled.
- End-to-end tests for each domain flow.
- Load test for high-volume schedule and webhook bursts.
- Security tests for webhook spoofing, expired token, unauthorized access, evidence upload, and data leakage.

21. Definition of Done

Area	Done When
Core	No hardcoded insurer/unit schedule; rules configurable and versioned.

Area	Done When
Reliability	Duplicate event/callback does not produce duplicate message or duplicate payment/update.
Stop Behavior	Completed/cancelled/paid state stops pending reminders within agreed SLA.
WA	Send + delivery + read + response + failure visible end-to-end.
Response	Structured CTA updates journey or creates correct task.
Audit	All manual and system transitions traceable by correlation ID.
Security	RBAC, secure webhook, token expiry, data minimization, attachment controls.
Operations	DLQ, replay, queue health, provider health, alerting available.
Documentation	OpenAPI, event catalog, config guide, runbook, deployment and rollback documented.
Testing	Unit, contract, integration, UAT, and load tests pass for agreed scope.

22. Minimum UAT Scenarios

ID	Scenario	Expected Result
UAT-01	Control H-1	Appointment aktif menghasilkan reminder tepat waktu dan pasien Confirm mengubah journey.
UAT-02	Reschedule	Reschedule membatalkan reminder lama dan membuat jadwal baru.
UAT-03	Lab completed	Specimen received menghentikan reminder berikutnya.
UAT-04	Radiology no-show	No-show menghasilkan follow-up dan worklist.
UAT-05	Medication complaint	Ada Keluhan membuat task klinis prioritas tinggi.
UAT-06	Invoice paid before send	Queued payment reminder tidak terkirim setelah callback paid.
UAT-07	Partial payment	Next reminder memakai outstanding terbaru.
UAT-08	WA failure	Permanent failure membuat task verifikasi kontak tanpa retry tak terbatas.

ID	Scenario	Expected Result
UAT-09	Duplicate event	Event replay tidak membuat reminder ganda.
UAT-10	Patient-submitted reimbursement	Patient receives document reminder until proof verified.
UAT-11	Hospital-submitted reimbursement	Only internal officer receives document SLA reminder.
UAT-12	Paid-to-patient	Payment-to-hospital reminder starts; patient can choose insurer not paid.
UAT-13	Payment proof	Upload proof creates verification task and secure evidence reference.
UAT-14	Out-of-order callback	READ state does not regress when DELIVERED callback arrives late.
UAT-15	Opt-out	Non-mandatory category is suppressed and reason logged.
UAT-16	Rule version	Existing reminder keeps original rule context; new event uses new effective version.
UAT-17	DLQ replay	Failed technical event can be replayed safely after fix.
UAT-18	RBAC	Unauthorized role cannot view/change restricted clinical/financial data.
UAT-19	Dashboard reconciliation	Dashboard totals match underlying reminder instances and payments.
UAT-20	Load	Agreed volume processed without lost or duplicate reminders.

23. Sample Message Templates

Template Code	Example
CONTROL_H1	Halo {{patient_name}}, pengingat kontrol Anda dengan {{doctor_name}} pada {{appointment_datetime}} di {{location}}. Pilih Konfirmasi Hadir atau Jadwal Ulang.
LAB_H1	Halo {{patient_name}}, pemeriksaan laboratorium Anda dijadwalkan {{appointment_datetime}}. Persiapan: {{preparation_instruction}}. Jika ada kendala, pilih Butuh Bantuan.
MEDICATION	Pengingat obat: {{medication_name}} {{dose_text}} pada {{schedule_time}} sesuai instruksi dokter. Pilih Sudah Diminum, Belum Diminum, atau Ada Keluhan.

Template Code	Example
PAYMENT_OVERDUE	Halo {{patient_name}}, invoice {{invoice_number}} memiliki sisa tagihan {{outstanding_amount}}. Silakan melakukan pembayaran melalui {{payment_link}} atau pilih Butuh Bantuan.
REIMB_DOC_PATIENT	Halo {{patient_name}}, dokumen reimbursement {{insurance_name}} perlu dikirim oleh pasien. Setelah dikirim, mohon unggah bukti melalui {{upload_link}}.
REIMB_SETTLEMENT	Halo {{patient_name}}, sesuai proses reimbursement {{insurance_name}}, kewajiban pembayaran ke rumah sakit untuk invoice {{invoice_number}} adalah {{outstanding_amount}}. Silakan bayar melalui {{payment_link}}. Jika dana asuransi belum diterima, pilih Belum Menerima Dana.

24. Handoff Checklist untuk Tim Developer

- Finalisasi event source dan owner tiap integration.
- Finalisasi provider WhatsApp dan template approval workflow.
- Finalisasi quiet hours, consent policy, dan message frequency cap.
- Finalisasi default reminder timing per unit; tetap configurable.
- Finalisasi role matrix dan akses data klinis/finansial.
- Finalisasi SLA task dan escalation chain.
- Finalisasi payment gateway/deep link dan evidence upload storage.
- Finalisasi reimbursement master per insurer: Hospital, Patient, Hybrid.
- Setujui API/event schema sebelum implementasi domain adapters.
- Setujui UAT dataset dan Definition of Done sebelum sprint integration.

Recommended MVP

Untuk rilis pertama, bangun core engine + WhatsApp + Appointment/Kontrol + Payment + Reimbursement. Setelah state model dan event pipeline stabil, tambahkan Lab/Radiology dan Medication. Urutan ini meminimalkan risiko membangun logika reminder terpisah di setiap modul.